

Exercise 2: Implementing Dependency Injection

Since in Exercise 1, we have already created the Library Management Application, so for this exercise, we will only update the files that we created in Exercise 1.

BookRepository.java:

```
package com.library.repository;

public class BookRepository {
    public void findAll() {
        System.out.println("Fetching all books from the database...");
    }
}
```

BookService.java:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {
    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void listBooks() {
        System.out.println("BookService: Delegating to Repository...");
        bookRepository.findAll();
    }
}
```

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
       http://www.springframework.org/schema/beans/spring-beans.xsd">
  <bean id="bookRepo" class="com.library.repository.BookRepository" />

  <bean id="bookService" class="com.library.service.BookService">
    <property name="bookRepository" ref="bookRepo" />
  </bean>

</beans>
```

App.java

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class App {
    public static void main(String[] args) {
        ApplicationContext context = new
        ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service = (BookService) context.getBean("bookService");
        service.listBooks();
    }
}
```

Output:

```
● adityahargunani@Adityas-MacBook librarymanagement % mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:librarymanagement >-----
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.1.0:clean (default-clean) @ librarymanagement ---
[INFO] Deleting /Users/adityahargunani/Downloads/Exercise 2- Implementing Dependency Injection/librarymanagement/target
[INFO]
[INFO] --- resources:3.0.2:resources (default-resources) @ librarymanagement ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- compiler:3.8.0:compile (default-compile) @ librarymanagement ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 4 source files to /Users/adityahargunani/Downloads/Exercise 2- Implementing Dependency Injection/librarymanagement/target/classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 0.563 s
[INFO] Finished at: 2026-06-24T18:49:04+05:30
[INFO]
● adityahargunani@Adityas-MacBook librarymanagement % mvn exec:java -Dexec.mainClass="com.library.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:librarymanagement >-----
[INFO] Building librarymanagement 1.4
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec:3.6.3:java (default-cli) @ librarymanagement ---
[INFO] BookService: Delegating to Repository...
[INFO] BookRepository: Finding all books in the database...
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 0.320 s
[INFO] Finished at: 2026-06-24T18:49:09+05:30
[INFO]
○ adityahargunani@Adityas-MacBook librarymanagement %
```